

Trained in chemistry and biology, Louis Pasteur was instrumental in establishing the field of microbiology. He discovered that microorganisms cause infectious disease and developed the first vaccines for rabies and anthrax. He also invented a method of sterilizing food, known as “pasteurization.”

MILESTONES

SENIOR SCIENTIST

In 1857, returns to his old college, the Ecole Normale Supérieure, as its Director of Science.

KEY PUBLICATION

Publishes his work on alcoholic fermentation in 1858, which leads to the invention of pasteurization.

PRIZE-WINNING

Wins 2,500 francs from the French Academy of Sciences in 1862, for disproving “abiogenesis.”

RABIES VACCINE

Performs the first rabies vaccination in 1885, on a boy who had been bitten by a dog with rabies.

ONGOING RESEARCH

Founds the Pasteur Institute in 1887, for research into the treatment of disease, which continues to this day.

Louis Pasteur was born and raised in a small town in the Jura region of France. His father, like generations before him, was a leather tanner. A talented artist, Louis spent much of his time drawing portraits in pastels. On his second attempt, he passed the entrance exams for an elite college in Paris, and in 1847 submitted PhD theses in both physics and chemistry. The next year, he became professor of chemistry at the University of Strasbourg and made his first scientific discovery: that some molecules exist in two mirror-image forms.

Fermentation, pasteurization, and microbes

In 1854, Pasteur became Dean of Sciences at the University of Lille. He became interested in a problem that was afflicting winemakers: the fermentation process often went wrong, spoiling the wine. Pasteur discovered that fermentation is caused by living yeast microbes and that souring occurs when the wine is contaminated by bacteria. In 1865, he patented a method for preventing such “diseases” of wine. The liquid was heated briefly to 140–212°F (60–100°C), which killed off the bacteria without affecting the taste. This process, which also worked on beer and milk, became known as pasteurization.

Pasteur’s research went further than solving the wine problem. He also answered the question of how microbial contamination occurs. In the 19th century, most scientists believed in the concept of

News spread quickly of the first successful rabies vaccination, administered to the 9-year-old Joseph Meister, and people soon flocked to Pasteur for treatment.

